

Lightweight U-Net Image Dehazing Network for Remote Airport Tower Control

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Abstract—Image dehazing is a significant part of low-level vision tasks in the area of airport traffic surveillance. However, the existing image dehazing methods based on deep learning suffer from the limitation on both model generalization and computational cost. A lightweight U-Net dehazing network based on improved inverted residual block (IIRB) and adaptive detail enhancement (ADE) module is proposed for remote airport tower control. Firstly, the low and mid-frequency feature was extracted with IIRB module from image smooth regions. Secondly, the high-frequency details and textures were extracted with ADE module, which further led to the enhancement of the network's expression in sky regions and detail features. Finally, the model was trained combining L1 loss and contrastive loss, which aimed to improve the model's generalization. To evaluate the effectiveness and practicality of the proposed method, the experiments were conducted both on SOTS dataset and real samples of airport hazy images. The experimental results show that, compared to lightweight AOD and weakly supervised RefineD, the proposed method achieves a balance between visual quality and real-time performance.

Index Terms—image dehazing, lightweight, inverted residual block, detail enhancement, contrastive loss, remote airport tower

I. INTRODUCTION

With the growing development of remote airport tower, the digitalization challenges of the traditional visual observation tower have been effectively addressed, substantially reducing airport operating costs [1]. However, the haze and other similar weather conditions considerably impair the monitoring quality of airport scenes, which potentially compromises the effectiveness and reliability of airport management systems.

Image dehazing aims to estimate potentially clear images from hazy images. Recently, the image dehazing based on deep learning has gained significant progress. Li et al [2] proposed the lightweight AOD on the basis of reconstructed atmospheric scattering model [3], which precisely improved the efficiency of image processing. Zhao et al [4] proposed the weakly-supervised RefineD, which fully integrated the dark channel prior and perceptual fusion strategy to reduce the dependence on the paired samples. In the field of remote airport tower image dehazing task, the lightweight model is more aligned with the practical needs due to the high-resolution of captured images. Therefore, to improve the robustness and accuracy of the method under the premise of ensuring the real-time

performance is a major problem to be solved in the processing of the airport hazy images.

II. APPROACH

A. Remote Airport Tower

Remote airport tower, which is usually applied in remote branch airports scenarios, obtains pictures from multiple high-definition cameras to extract key frames for processing based on image enhancement methods [5].



Fig. 1. A sample image of remote airport tower monitoring

As illustrated in Fig. 1, the captured image is typically stitched from multiple ultra-high resolution images, which places high demands on the processing quality and efficiency.

B. Overall Architecture

A lightweight U-Net dehazing network based on IIRB and ADE module is proposed to meet the requirements both on image processing quality and speed, as presented in Fig. 2.

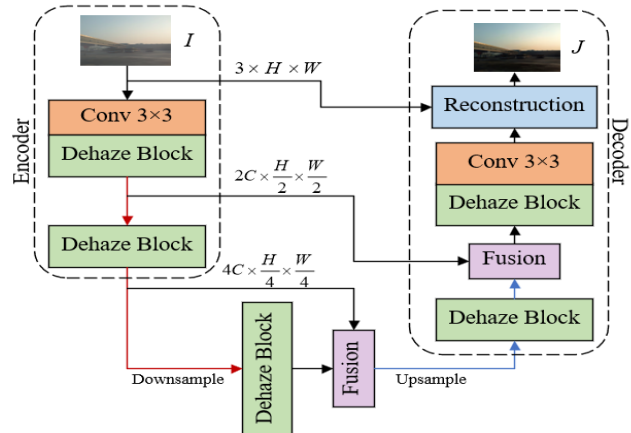


Fig. 2. The proposed method architecture

For a hazy image $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width and channel of the input image respectively. Specifically, shallow features are extracted by the embedding layer, and then followed by two symmetric encoder-decoder structure combined with down-sample and up-sample, to capture the high-level features. The center bottleneck is designed as dehazing module as well. Benefiting from the symmetric encoder-decoder structure, the model realizes the dynamic fusion of features from different branches through skip connections to recover the image details.

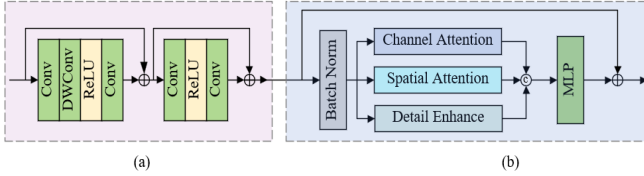


Fig. 3. Dehaze block: (a)IIRB; (b)ADE

C. Improved Inverted Residual Block

Most of the dehazing methods are realized by increasing the depth and the width of the network [6], which result in more computational cost and a decrease in the processing speed. As depicted in Fig. 3(a), on the basis structure of IRB [7] module, the proposed IIRB module demonstrates significant advantages in processing low and mid-frequency information of the image, while reducing computational cost.

Specifically, the low-dimensional input is converted into a high-dimensional representation by point-wise convolution. Then, the low-frequency information is extracted by depth-wise convolution, which performs depth convolution and point-wise convolution on each channel separately. Finally, the high-dimensional features are projected to the original input by the nonlinear layer and point-by-point convolution layer so as to achieve feature fusion across various levels.

In order to obtain more detailed image features as well as to reduce color distortion and artifacts in the sky region, feature enhancement is applied to the output of IRB module. The superimposed 1×1 convolution layer is utilized to strengthen the token representation, while the activation layer is able to enhance the model representation. Additionally, the skip connection is adopted to finely fuse the features and alleviate the problem of vanishing gradient.

D. Adaptive Detail Enhancement Module

Extracting more high-frequency information facilitates the restoration of the edges and texture structures of the image. The proposed ADE module, shown in Fig. 3(b), leverages attention mechanisms to fully extract high-frequency features, which improves the ability to process complex textures and recover details. The atmospheric scattering model [3] describes the degradation process of hazy images, which is formulated in Eq.(1):

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (1)$$

Where $I(x)$ is the hazy image; $J(x)$ is the clear image; atmospheric light A serves as a shared global variable, and $t(x)$ is a position-dependent local variable.

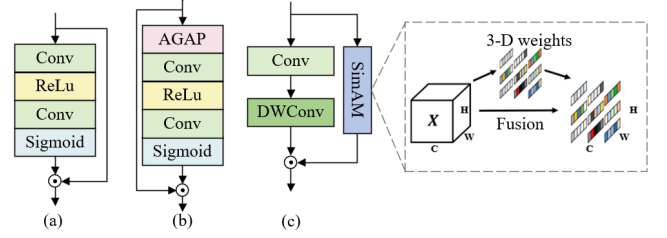


Fig. 4. The structure of attention mechanisms: (a) Pixel attention; (b) Channel attention; (c) Detail enhancement

Considering the uneven distribution of haze, the pixel attention (PA) is used to dynamically adjust the degree of attention to different regions. The local regions are weighted so as to improve the encoding accuracy of the value of $t(x)$. As demonstrated in Fig. 4(a), PA module compresses the channel dimensions of the input feature map, generating two-dimensional attention weight $1 \times H \times W$, which is subsequently multiplied element-wise with the initial feature map.

The channel attention (CA) module aims to adjust the weight of each channel, which effectively allocates higher weights to key channels while filtering out irrelevant information and noise. CA module could facilitate the extraction of shared global information and encoding of the atmospheric light A . In Fig. 4(b), the adaptive global average pooling (AGAP) is first employed to compress the global information into channel descriptors, and the output is presented in Eq.(2):

$$p_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j) \quad (2)$$

where $X_c(i, j)$ represents the value of c -th channel X_c at the position (i, j) . Then the output generates one-dimensional weight $C \times 1 \times 1$ through the activation layer.

CA and PA module generate one-dimensional and two-dimensional attention weights respectively, which restricts the ability to learn three-dimensional attention weights that vary across both channel and space. In order to compensate for the crucial features that might have been overlooked in the feature extraction process, the dual-branch detail enhancement (DE) module is proposed, as shown in Fig. 4(c). Specifically, SimAM [8] is introduced in the global branch to generate lightweight three-dimensional attention weights, and the output is defined as in Eq.(3):

$$F_{Sim} = \sigma\left(\frac{1}{E}\right) \odot x_{in} \quad (3)$$

Where E denotes the minimum energy summary of neurons for all channels and spatial dimensions.

In the local branch, the high-frequency feature is extracted via a 3×3 depth-wise convolution. The outputs of three parallel modules are integrated along the channel dimension, and then sent to the multi-layer perception (MLP) layer, where

ghost convolution [9] is utilized to reduce the computational cost.

E. Loss Function

The proposed model is trained by a combination of L1 loss and contrastive loss, where the contrastive loss extracts features by pre-training visual geometry group network-19 (VGG19) [10] to compare the similarity and difference in the feature space. The joint loss function is defined in Eq.(4):

$$L = \alpha \|J - \hat{J}\|_1 + \beta \sum_{i=0}^n \omega_i \cdot \frac{\|F_i(J) - F_i(\hat{J})\|}{\|F_i(I) - F_i(J)\| + \varepsilon} \quad (4)$$

Where $\hat{J}$ is a restored image; ω_i is the weight coefficients; F_i is the feature extracted from the i -th layer of pre-training VGG19; ε aims to prevent the de-zeroing from being set incorrectly; α and β are the hyperparameters that balance L1 loss and contrastive loss, which set to 1 and 0.5 sequentially.

III. EXPERIMENTS AND ANALYSIS

To validate the effectiveness and practicality of the proposed method, experiments are conducted on the standard SOTS dataset and real samples of airport hazy images. The experimental dataset and parameter settings, as well as the result analysis of the proposed method are elaborated in this section.

A. Dataset and Experiment Settings

Considering the lack of datasets related to remote airport tower, 14000 images are selected from the ITS and OTS subsets of the RESIDE [11] dataset for training. The SOTS subset and real samples of airport hazy images are used for testing. The parameters of the experimental environment are depicted in Tab. I.

TABLE I
PARAMETERS OF THE SOFTWARE AND HARDWARE

Parameters	Value
CPU	Intel Xeon Gold 6330
GPU	NVIDIA RTX 3090
OS	ubuntu 20.04
Pytorch	1.11.1

During the training process, images are randomly cropped to 256×256 pixels and data enhancement is achieved by horizontal flipping and angular random rotation. The specific hyperparameter settings are shown in Tab. II

TABLE II
PARAMETERS OF THE TRAINING SETTINGS

Parameters	Value
Optimizer	AdamW
Learning rate	0.0002
Decay rate β_1	0.9
Decay rate β_2	0.999
Batch size	16
Epochs	300

B. Evaluation Metrics And Result Analysis

Comparisons and result analysis are conducted between the proposed method and two benchmark methods, the lightweight AOD and the weakly supervised RefineD, both on SOTS dataset and real samples of airport hazy images.

1) *Qualitative Analysis*: Four samples of the visual results are selected to analyze the utility and reliability of each method. In Fig. 5(b), it is observed that AOD suffers from the excessive reduction of brightness in img1, which is attributed to its partial reliance on the atmospheric scattering model. The red box in Fig. 5(c) shows that RefineD results in different degrees of color distortion in all samples with noticeable blocky artifacts particularly in the sky region of img4. In contrast, the proposed method generates natural restoration with superior ability to reduce color distortion and artifacts in the sky region, while preserving more airport road sign details and edge information.

2) *Quantitative Analysis*: On the basis of paired images, Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) are selected as the referenced evaluation metrics for the SOTS dataset. As presented in Tab. III, the proposed method has an obvious advantage in the objective evaluation indicators, which receives with 31.57dB PSNR and 0.974 SSIM in SOTS dataset, achieving better processing quality with fewer computational cost.

TABLE III
QUANTITATIVE COMPARISON RESULTS ON SOTS DATASET

Method	PSNR	SSIM	#Param	GFLOPs
AOD	20.29	0.876	0.002M	0.11
RefineD	24.23	0.943	65.80M	75.41
Proposed	31.57	0.974	0.90M	9.32

Due to the lack of clear images corresponding to the real samples of airport hazy images, Information Entropy (IE) and Variance (VAR) are adopted as the no-reference evaluation metrics [12], while IE measures the amount of image information and VAR evaluates the degree of image gray scale dispersion.

TABLE IV
QUANTITATIVE COMPARISON RESULTS ON REAL AIRPORT HAZY IMAGE SAMPLES

Samples	IE VAR		
	AOD	RefineD	Proposed
img1	6.58 1517.3	6.63 2478.5	6.84 2147.7
img2	6.88 1943.4	7.16 1939.2	7.15 3387.8
img3	6.30 5329.2	6.65 5113.3	7.05 5329.3
img4	6.80 2715.8	7.07 2505.0	7.24 2904.1

As presented in Tab. IV, the objective evaluation metrics of the proposed method are superior to the comparison methods on the whole. Compared to RefineD, the variance of img2 increases by 74.7%, indicating satisfactory contrast and visual quality. Meanwhile, compared to AOD, the value of IE in img3 rises by 11.9%, reflecting a extensive detailed information.

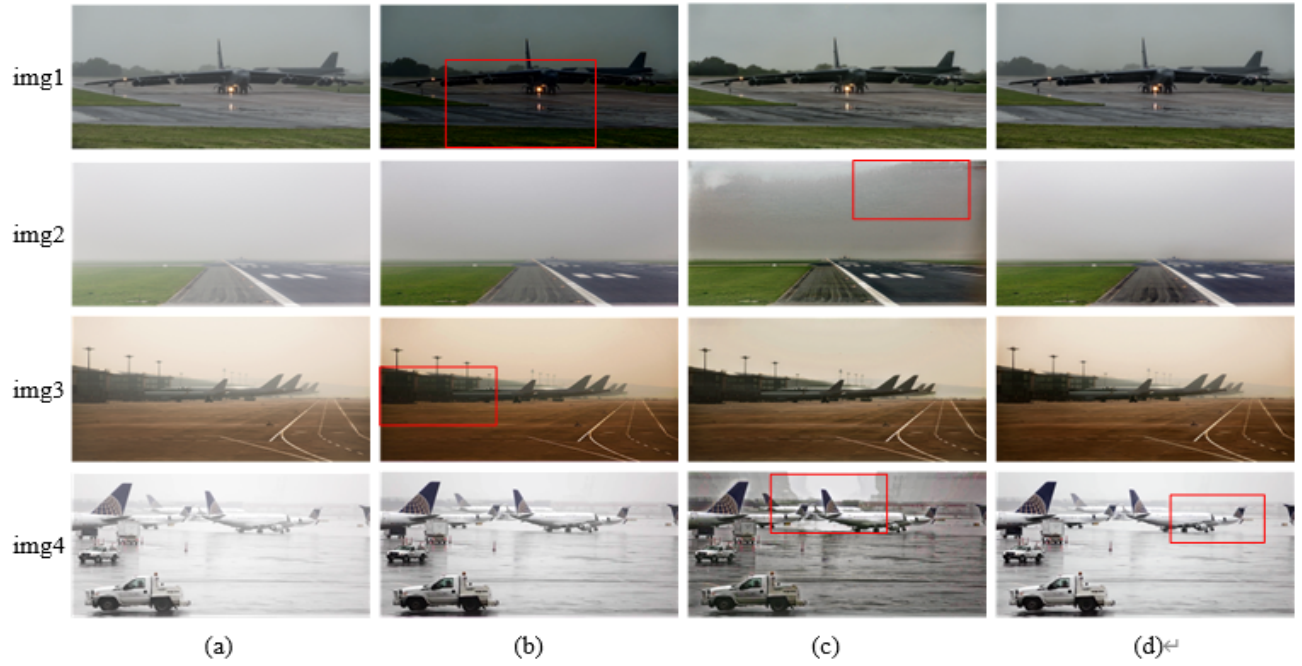


Fig. 5. Comparison of dehazing results on real samples of airport hazy images: (a) Hazy; (b) AOD; (c) RefineD; (d) Proposed

To summarize, the proposed method effectively recover the valuable information with better robustness.

To prove the advantage of the proposed method in terms of computational efficiency, a comparison on running time is performed with the objective indicator. As shown in Tab. V, compared to AOD, the proposed method achieves a 34% increase in processing speed, which effectively confirms the superiority in real-time image processing for remote airport tower.

TABLE V
RUNNING TIME (S)

Samples	AOD	RefineD	Proposed
img1	0.29	2.73	0.19
img2	0.12	3.21	0.15
img3	0.25	3.22	0.11
img4	0.34	4.24	0.21
average	0.25	3.35	0.16

C. Ablation experiments

The ablation experiments are conducted to demonstrate the rationality of the dehazing module design, while the settings are consistent with the model training.

To verify the impact of IIRB module on the overall network performance, three sets of ablation experiments are conducted by varying the number of IIRB blocks. As depicted in Tab. VI, the results demonstrate that the proposed method achieves an improvement of 2.33 dB in PSNR with a 25.9% increase in computational cost while using two blocks. However, further increasing blocks leads to the reduction of PSNR gain and

growth of cumulative computational cost. To meet the real-time requirement, the network are optimal when 2 IIRB modules are used.

TABLE VI
ABLATION ANALYSIS ON IIRB MODULE

Num. block	PSNR	GFLOPs
1	29.24	7.40
2	31.57	9.32
3	31.98	11.2

For the purpose of verifying whether Detail Enhancement (DE) module is more suitable for image dehazing, three sets of comparison experiments are conducted to compare the effects. BL denotes the baseline model in which the dehazing module consists of two IIRB modules and the MLP layer. BL+CA+PA denotes the addition of channel and pixel attention on the basis of the baseline; BL+CA+PA+DE denotes the addition of channel, pixel and DE modules sequentially.

TABLE VII
ABLATION ANALYSIS ON DE MODULE

Method	PSNR	SSIM
BL	29.32	0.957
BL+CA+PA	30.01	0.962
BL+CA+PA+DE	31.57	0.974

As illustrated in Tab. VII that, incorporating DE module leads to an improvement of 1.56 dB PSNR and 0.012 SSIM. It is believed that the main reason for the high effectiveness is that, DE module, as a supplementary enhancement module,

extracts high-frequency information and improves the model representation and dehazing performance.

IV. CONCLUSION

In this paper, a lightweight U-Net dehazing network is proposed for the task of remote airport tower image dehazing. Specifically, the U-Net structure is employed as the main architecture of the proposed method. Then, a combination of IIRB module and ADE module is utilized to extract the low, medium and high-frequency information of the image adequately, which fully enhances model representation and the ability of noise removal. The experimental results indicate that the proposed method achieves a balance between visual quality and real-time performance. Compared to lightweight AOD and weakly supervised RefineD, the proposed method demonstrates an effective reduction in color distortion and artifacts, particularly in the sky regions of hazy airport images, while successfully restoring detailed information. In terms of image processing speed, the proposed method achieves a 34% increase in processing speed over the lightweight AOD, which effectively exhibits potential to fulfill the practical requirements of robustness and real-time performance necessary for airport traffic surveillance.

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